有理数类

```C++
#include <iostream>
#include <algorithm> // 包含算法库，用于求最大公约数

using namespace std;

class Rational {
private:
    int numerator;   // 分子
    int denominator; // 分母

public:
    // 默认构造函数，带默认值，简化分数形式
    Rational(int num = 0, int denom = 1) {
        int gcd = __gcd(num, denom); // 使用 __gcd 函数求最大公约数
        numerator = num / gcd;
        denominator = denom / gcd;
    }

    // 加法
    void add(const Rational &r1, const Rational &r2) {
        numerator = r1.numerator * r2.denominator + r2.numerator * r1.denominator;
        denominator = r1.denominator * r2.denominator;
        simplify();
    }

    // 减法
    void minus(const Rational &r1, const Rational &r2) {
        numerator = r1.numerator * r2.denominator - r2.numerator * r1.denominator;
        denominator = r1.denominator * r2.denominator;
        simplify();
    }

    // 乘法
    void multi(const Rational &r1, const Rational &r2) {
        numerator = r1.numerator * r2.numerator;
        denominator = r1.denominator * r2.denominator;
        simplify();
    }

    // 除法
    void divide(const Rational &r1, const Rational &r2) {
        numerator = r1.numerator * r2.denominator;
        denominator = r1.denominator * r2.numerator;
        simplify();
    }

    // 打印分数形式
    void printFormal() const {
        cout << numerator << "/" << denominator;
    }

    // 打印小数形式，限制小数点后两位
    void printDecimal() const {
        double result = static_cast<double>(numerator) / denominator;
        printf("%.2f", result); // 使用 printf 控制小数位数
    }

    // 打印分数和小数形式
    void printBoth() const {
```

```cpp
        cout << "Formal format: ";
        printFormal();
        cout << ", Decimal format: ";
        printDecimal();
        cout << endl;
    }

private:
    // 简化分数
    void simplify() {
        int gcd = __gcd(numerator, denominator); // 使用 __gcd 函数求最大公约数
        numerator /= gcd;
        denominator /= gcd;
    }
};

//StudybarCommentBegin
int main()
{
    int firstN,firstD,secondN,secondD;
    char op;

//    cout<<"Please input the numerator of first Rational: ";
    cin>>firstN;
//    cout<<"Please input the denominator of first Rational: ";
    cin>>firstD;
//    cout<<"Please input the numerator of second Rational: ";
    cin>>secondN;
//    cout<<"Please input the denominator of second Rational: ";
    cin>>secondD;


    Rational r1(firstN,firstD),r2(secondN,secondD),r3;


    cin>>op;

    cout<<"the Formal format of the first rational is : ";
    r1.printFormal();
    cout<<"the Formal format of the second rational is : ";
    r2.printFormal();
    cout<<endl;

    switch(op)
    {
        case '+' :
            r3.add(r1,r2);
            r3.printBoth();
            break;
        case '-' :
            r3.minus(r1,r2);
            r3.printBoth();
            break;
        case '*':
            r3.multi(r1,r2);
            r3.printBoth();
            break;
        case '/':
            r3.divide(r1,r2);
            r3.printBoth();
            break;
        default:
            cout<<"Invalid operator!";
    }

    return 0;
}
//StudybarCommentEnd
```